

the impaired around the world. Going one step further, the Monash Vision Group is developing Gennaris—a bionic vision system that bypasses damage to the eye and optic nerve to restore functional vision for people who have injured both these or lost sight due to glaucoma and acquired retinal disease. This system interfaces directly with the brain, bypassing the retina and optic nerve.

Elsewhere, researchers are also exploring biocompatible, implantable photonic devices that can improve health monitoring, diagnostics and light-activated therapies. Consider the possibility of biocompatible and wirelessly-powered light-emitting diodes (LEDs) and miniature lasers implanted inside the body. Advances in biotechnology, such as optogenetics, will enable these photonic implants to be integrated tightly with neurological or physiological circuits.

A good interface between implanted devices and the brain can help in great ways—for example, it can return motor function to amputees and people paralysed due to stroke or spinal cord injury. A minimally invasive electrode called the Stentrode developed at the University of Melbourne might be a step in this direction. Implanted into a blood vessel adjoining the area of the brain that controls movement, it may help control an exoskeleton, enabling crippled or paralysed people to move. The implant can apparently be installed without opening the skull, which is what makes it attractive!

Neuralink, a company funded by Elon Musk, is also working on implantable brain-computer interfaces. They are developing syringe-injectable, flexible, sub-micron-thickness substrates that can be used in implantable electronics. The substrate is soft enough to sit harmlessly in the brain, and has electrical properties that enable only the targeted part of the brain to receive the electrical stimulus.

Brain-computer interface is the future of implantable systems. It can



A range of ingestible cameras, which can help doctors to see the patients' oesophagus, small intestine and colon (Credit: Given Imaging)

help people with degenerative brain diseases and neurological disorders. However, it must be handled with care because an electrode implanted in the brain can be used both for good and bad purposes!

Rise of edibles for diagnostics and drug delivery

When the electronic device needs to stay inside the body forever or for a reasonably long time, it is worth operating on a patient to implant the device. However, if you just want it to stay inside for a few minutes, hours or even days, for the purpose of monitoring a health condition or temporarily dispensing some medicines, operating on the patient doesn't make sense. This requirement led to the development of ingestible or edible electronics, which industry-watchers expect to create huge waves like wearables did.

The electronics that you swallow, encapsulated in a pill, will sit in your gastrointestinal tract for a short time, before being ejected from your body like regular food waste. During this time it can capture videos, release drugs, monitor heart rate and respiration, and perform other such tasks.

Proteus Digital Health was one of the pioneers in ingestible tech, and their technology is now used commercially by close to ten health systems. Their solution comprises a pill, a patch that is attached outside to the side of the stomach, and a mobile

app/desktop portal. The pill is made of whatever drugs are required, and fitted with a sensor made of natural, ingestible materials like copper, magnesium and silicon. When a patient swallows the pill, it dissolves like a normal pill in the stomach but leaves behind a sensor, which is activated by fluids in the body. This sensor sends a signal to the patch, which also measures heart rate, body position and time of medication detection. This information is sent to the patient's or doctor's mobile phone.

Regular drug intake is very important for those undergoing complex medical treatments such as organ transplants. This pill could be very useful for such patients. Proteus has teamed up with Tokyo-based firm Otsuka to embed Proteus' sensors into Abilify—a drug used for serious mental illnesses.

Another forerunner in the space is Israel-based Given Imaging. Their PillCam series comprises pills with ingestible cameras, which can help doctors to view different parts of the patient's digestive system like the oesophagus or colon. It is a painless alternative to tests like endoscopy and colonoscopy. The PillCam Colon, for instance, uses a battery-powered camera to take high-speed photos as it slowly goes down the intestinal tract over a time period of eight hours. The images are transmitted to a recording device worn around the patient's waist and later reviewed by a doctor. Although